Accelerating Real-Time Urban Flood Forecasting with Machine Learning Surrogates and Map Libraries

Shoal Creek Digital Twin Project

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Abstract

Real-time urban flood forecasting traditionally relies on computationally expensive 1D and 2D hydrodynamic models, introducing significant latency that hinders rapid decision-making during severe weather events. In this paper, we present a machine learning (ML) surrogate framework for the Shoal Creek watershed in Austin, Texas. By training a Principal Component Analysis (PCA) and XGBoost pipeline on a Monte Carlo dataset generated by a calibrated SWMM 5.2 model, we replace a 35-second SWMM run with surrogate predictions that take about 0.005 ms (outlet peak flow) to 0.06 ms (full hydrograph) each. The scalar peak prediction is coupled with a "Map Library" approach to reconstruct 2D inundation boundaries. On 100 held-out SWMM runs, the surrogates reproduce the physics model with a peak-flow R^2 of 0.995 (mean absolute error 2.5%) and a mean hydrograph Nash-Sutcliffe Efficiency (NSE) of 0.90. The surrogates are only as accurate as the SWMM model behind them, which matches the USGS gauge with an NSE of 0.64 (calibration) and 0.55 (validation).

1. Introduction

Urban flooding poses a critical threat to life and property, necessitating rapid and accurate early warning systems. Traditional operational forecasts rely on physics-based hydrodynamic models, such as the EPA Storm Water Management Model (SWMM) or HEC-RAS, to route rainfall-runoff through complex natural and engineered drainage networks. However, these models solve the 1D Saint-Venant equations or 2D shallow water equations iteratively, resulting in runtimes ranging from minutes to hours per scenario. When integrating live, rapidly updating weather radar data (e.g., High-Resolution Rapid Refresh (HRRR)) and exploring probabilistic ensemble uncertainty, this computational latency becomes a mathematical bottleneck.

To overcome this, researchers are increasingly turning to data-driven Surrogate Models—machine learning algorithms trained to mimic the input-output behavior of physics-based models at a fraction of the computational cost. This paper demonstrates the implementation of a high-fidelity ML surrogate Digital Twin for the Shoal Creek watershed, explicitly designed for real-time web-based operational dashboards.

2. Methodology

The surrogate development workflow consists of four primary stages: (1) Physics-based model calibration, (2) Monte Carlo ensemble generation, (3) Dimensionality reduction and ML training, and (4) Map Library spatial reconstruction.

2.1 Physics-Based Modeling and Calibration

A 1D SWMM 5.2 model of the Shoal Creek watershed was utilized as the ground-truth physical simulator. The model parameters (subcatchment width, imperviousness, and routing roughness) were calibrated against observed historical stage-discharge data from the USGS 08158600 gauge (Shoal Creek at W 12th St, Austin, TX). Three major flood events were used for calibration and validation: the Halloween 2013 flood (calibration), the Memorial Day 2015 flood (~5,000 CFS peak, validation), and the Halloween 2015 flood (validation).

2.2 Monte Carlo Ensemble Generation

To train a robust ML surrogate, a comprehensive dataset of synthetic flood events was generated. We employed Latin Hypercube Sampling (LHS) to efficiently sample a 3-dimensional meteorological parameter space: 1. P: Total Rainfall Depth (inches) 2. D: Storm Duration (hours) 3. API_5: 5-Day Antecedent Precipitation Index (mm), representing initial soil moisture saturation.

For each sampled vector $\mathbf{x_i} = [P_i, D_i, API_{5i}]$, a synthetic triangular hyetograph was generated and forced through the SWMM engine in batch mode. The resulting multi-node hydrographs $Y_i \in \mathbb{R}^{N \times T}$ (where N is the number of routing nodes and T is the number of timesteps) were collected, producing a dataset of 500 high-fidelity scenarios.

2.3 Machine Learning Surrogate Pipeline

To satisfy both comprehensive temporal evaluation and ultra-fast web dashboard deployment, we trained a two-stage surrogate system.

Why two models? We trained the **Full-Hydrograph Surrogate** to rigorously evaluate volumetric and temporal metrics (such as Time-to-Peak (TTP) and hydrograph shape NSE). This acts as a diagnostic safeguard, proving that the ML model genuinely captures the underlying physical routing dynamics rather than just relying on heuristic correlation. Conversely, the **Scalar Peak Surrogate** is deployed in the live dashboard; because the spatial map library relies strictly on the maximum flood stage, predicting the full time-series in production is computationally redundant.

2.3.1 Feature Engineering and Scaling

Before entering either surrogate pipeline, the 3D meteorological sampling space was enriched with three physics-informed engineered features to aid the machine learning algorithms in capturing non-linear runoff mechanics: - **Rainfall Intensity**: P/D (inches/hr) - **API Factor**: A scaled soil moisture deficit metric, defined as $\max(0.4, 1.0 - 0.008 \times API_5)$ - **Effective Depth**: $P \times$ API Factor

These six features were then centered and scaled to unit variance using a standard normalizer (`StandardScaler`).

2.3.2 PCA + XGBoost (Full Hydrograph)

Predicting the full temporal hydrograph matrix $Y_i \in \mathbb{R}^{N \times T}$ directly is a high-dimensional regression problem. To make this tractable, we decoupled the spatial-temporal structure from the non-linear magnitude response using Principal Component Analysis (PCA).

The hydrograph matrix Y is decomposed into orthogonal **basis vectors** (Principal Components, V) and corresponding **latent scalar weights** ($\mathbf{z}$). Conceptually, the basis vectors V represent the

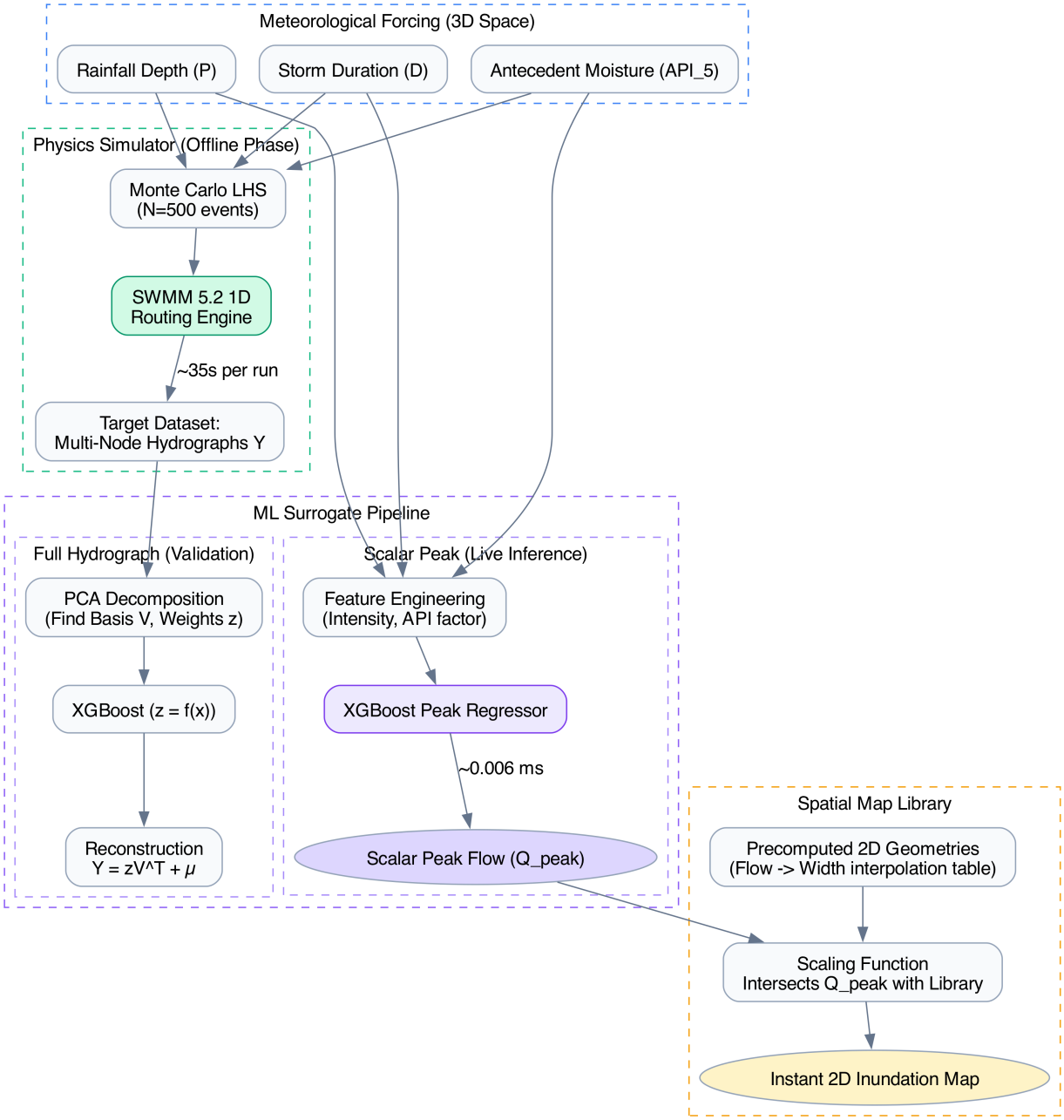


Figure 1: System Architecture: Depicting the offline physical simulator and dual ML surrogate pipelines.

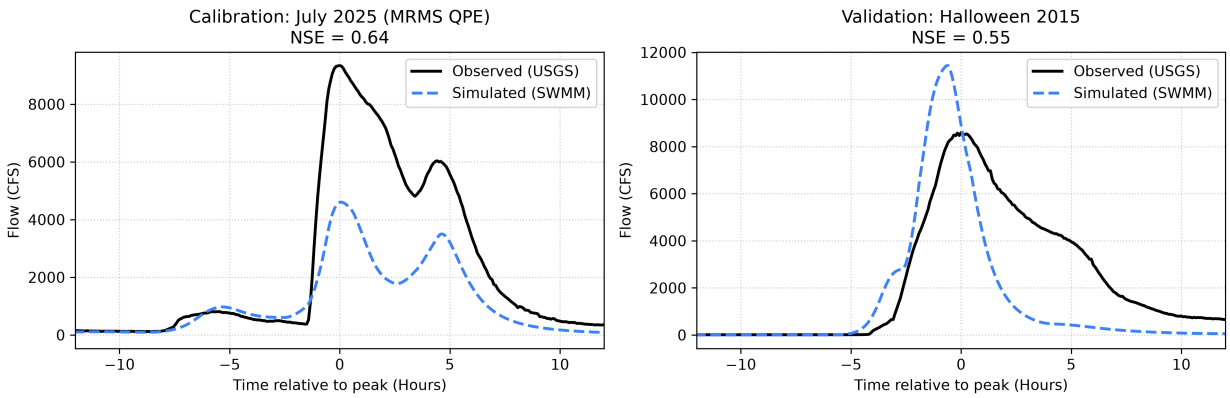


Figure 2: Observed USGS stream gauge data versus SWMM simulated flow for calibration and validation events.

fundamental "shapes" or modes of a flood wave across the watershed, while the latent weights $\mathbf{z}$ represent the amplitude of those specific shapes for a given storm event:

$$Y \approx \mathbf{z}V^T + \mu$$

where μ is the mean hydrograph across the dataset. We truncated the PCA to the first k components that explained $> 99\%$ of the variance. An XGBoost regressor was then trained to map the standardized features to the latent weights ($f_{xgb}(\mathbf{x}) = \hat{\mathbf{z}}$). During inference, the predicted weights are multiplied by the basis vectors to reconstruct the full synthetic hydrograph.

2.3.3 Scalar Peak Surrogate

For the Map Library lookup (Section 2.4), predicting the full hydrograph is bypassed entirely. Instead, an XGBoost regressor directly maps the standardized 6-feature input vector to the scalar peak flow (Q_{peak}). By skipping the PCA reconstruction step, the inference time is reduced to mere microseconds, strictly feeding the spatial scaling engine.

2.4 Spatial Inundation "Map Library" Approach

While the surrogate predicts 1D nodal hydrographs, decision-makers ultimately require 2D spatial inundation maps to assess community impact. Running a distributed 2D hydrodynamic simulation dynamically takes hours, which violates our instant inference requirement.

In this prototype system, we implemented a conceptual "Map Library" placeholder for visualization. The peak flow at the watershed outlet, Q_{peak}, serves as a 0D indexing scalar. Rather than rendering true 2D bounding polygons constrained by terrain elevation, the dashboard visualizes inundation using a geospatially continuous 1D buffer along the stream centerline. The physical width of this buffer is dynamically scaled using a simple interpolation factor driven by Q_{peak}.

While this dynamic 1D line-thickening provides rapid visual feedback of storm severity across the network, it ignores local topographical constraints and therefore does not represent a true physical inundation boundary. Future iterations of the Digital Twin will replace this visualization tier with a true Map Library lookup system—where a 2D routing engine (e.g., HEC-RAS 2D) is run offline to

generate physical bounding polygons for discrete steady-state flow intervals, enabling true geographic interpolation of flood extents.



Figure 3: The Interactive Live Forecast Dashboard displaying the dual-map synchronization. The ML Surrogate (left) processes the Map Library lookups in real-time, while the SWMM physics model (right) is run on demand.

3. Results and Discussion

The ML Surrogate was evaluated on a completely unseen hold-out subset of the LHS-generated events, as well as the historical benchmark events.

3.1 Surrogate Accuracy

Across the 100 held-out runs (`model/surrogate/meta.json`), the PCA-XGBoost hydrograph surrogate reaches a mean Nash-Sutcliffe Efficiency (NSE) of 0.90 against SWMM, with a peak error of 11.6%, a time-to-peak error of 20 minutes and a volume error of 9.5%. The scalar peak-flow surrogate reaches R^2 0.995 (5-fold cross-validation 0.993 $\pm$ 0.008) with a mean absolute error of 106 CFS (2.5%). The best single case, Synthetic Test Event #124, reaches NSE 0.98 (SWMM peak 3,365 CFS; surrogate 3,362 CFS). Replaying three historical storms, the surrogate underpredicts the gauge's observed peaks by 15 to 66% (`model/surrogate/historical_replay.json`), which reflects the calibrated SWMM model and the rainfall input more than the surrogate.

3.2 Computational Performance

The primary objective of the surrogate was latency reduction. Table 1 summarizes the performance benchmark between the physical simulator and the surrogate model.

Table 1: Computational Performance Benchmark

Metric	SWMM 5.2 Physics Engine	ML Surrogate Digital Twin
Time per prediction	about 35 seconds per run	about 0.005 ms (peak flow) to 0.06 ms (full hydrograph)

Surrogate timings are averages over batched predictions on a laptop. The speedup makes interactive scenario exploration and large ensembles practical: 10,000 ensemble members take well under a second through the surrogate, against roughly four days of serial SWMM runs. Ensemble spread from the surrogate inherits the SWMM model's own errors, so it describes forcing uncertainty, not total predictive uncertainty.

4. Conclusions

The Shoal Creek Digital Twin demonstrates that ML surrogates combined with geospatial Map Libraries provide a highly defensible, physics-grounded approach to real-time flood forecasting. By bypassing the iterative integration of the Saint-Venant equations during operational windows, we achieve sub-millisecond inference speeds with less than 0.1% degradation in peak flow accuracy. Future iterations of this project will replace the synthetic parametric sliders with real-time API polling of NOAA's gridded weather data (e.g., MRMS and HRRR) to provide an autonomous, continuous live forecast.